Decision Structures

TOPICS

- The `if` Statement
- The `if-else` Statement
- Comparing Strings
- Nested Decision Structures and the `if-elif-else` Statement
- Logical Operators
- Boolean Variables

THE IF STATEMENT

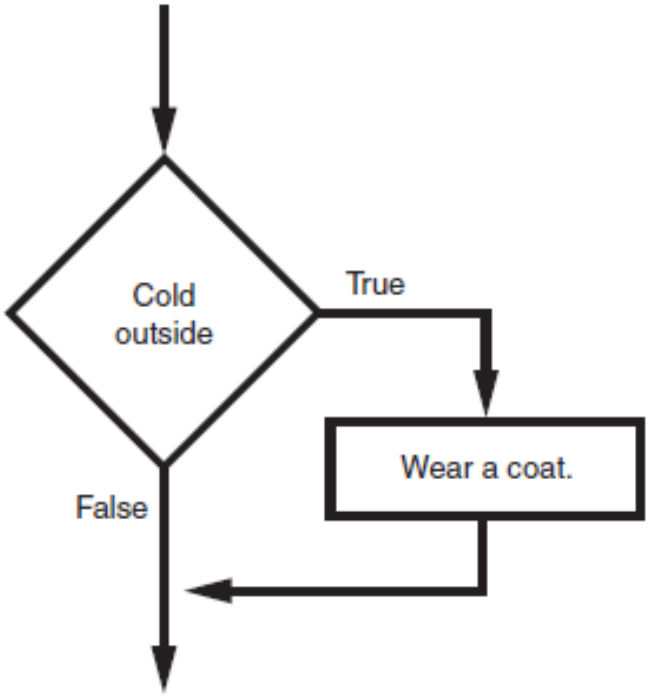
- Control structure: logical design that controls order in which set of statements execute
- Sequence structure: set of statements that execute in the order they appear
- Decision structure: specific action(s) performed only if a condition exists
- Also known as selection structure

THE IF STATEMENT (CONT'D.)

- In flowchart, diamond represents true/false condition that must be tested
- Actions can be *conditionally executed*
  - Performed only when a condition is true
- Single alternative decision structure: provides only one alternative path of execution
  - If condition is not true, exit the structure

THE IF STATEMENT (CONT'D.)

Figure 3-1 A simple decision structure



THE IF STATEMENT (CONT'D.)

- Python syntax:

```
if condition:
    Statement
    Statement
```

- First line known as the `if` clause
- Includes the keyword `if` followed by condition
- The condition can be true or false
- When the `if` statement executes, the condition is tested, and if it is true the block statements are executed. otherwise, block statements are skipped

MIRRORING GAME 1

- Player 1 reads a card with a number.
- Player 1 writes the number on the left side of the board.
- Player 2 reads the number written by player one on the board.
- If the number is odd: player 2 writes the double of the number on the right side of the board.
- Else if the number is even, player 2 writes the triple of the number of the right side of the board.

MIRRORING GAME 2

- Player 1 reads a card with a number or a word.
- Player 1 if there is a number he writes it on the left side of the board.
  - Else if there is a word. He writes nothing.
- Player 2 reads the number written by player one on the board.
- If the number is odd: player 2 writes the double of the number on the right side of the board.
  - Else if the number is even, player 2 writes the triple of the number of the right side of the board.

BOOLEAN EXPRESSIONS AND RELATIONAL OPERATORS

- Boolean expression: expression tested by if statement to determine if it is true or false
- Example: a > b
- `true` if a is greater than b; `false` otherwise

- Relational operator: determines whether a specific relationship exists between two values
- Example: greater than (>)

BOOLEAN EXPRESSIONS AND RELATIONAL OPERATORS (CONT'D.)

- >= and <= operators test more than one relationship
  - It is enough for one of the relationships to exist for the expression to be true
- == operator determines whether the two operands are equal to one another
  - Do not confuse with assignment operator (=)
- != operator determines whether the two operands are not equal

BOOLEAN EXPRESSIONS AND RELATIONAL OPERATORS (CONT'D.)

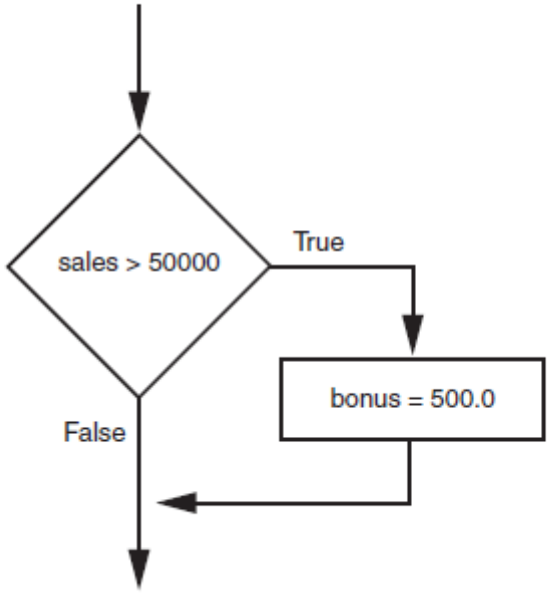
Table 3-2 Boolean expressions using relational operators

Expression	Meaning
`x > y`	Is x greater than y?
`x < y`	Is x less than y?
`x >= y`	Is x greater than or equal to y?
`x <= y`	Is x less than or equal to y?
`x == y`	Is x equal to y?
`x != y`	Is x not equal to y?

BOOLEAN EXPRESSIONS AND RELATIONAL OPERATORS (CONT'D.)

- Using a Boolean expression with the > relational operator

Figure 3-3 Example decision structure



BOOLEAN EXPRESSIONS AND RELATIONAL OPERATORS (CONT'D.)

- Any relational operator can be used in a decision block
  - Example: if balance == 0
  - Example: if payment != balance
- It is possible to have a block inside another block
  - Example: `if` statement inside a function
  - Statements in inner block must be indented with respect to the outer block

SUCCESSION OF IF STATEMENTS

- If the conditions are not mutually exclusive.
- Examples of non mutually exclusive attribute variables are:
  - Age and gender: female and more than 65.
  - Color and size: small blue.
- Examples of mutually exclusive attribute variables:
  - Large and small.
  - Hot/warm and cold.

CARD GAME EXAMPLE

- If a card is red you win one point
- If card is larger than five you win five points.
- If the card is a heart you win ten points
- If the card is a diamond you win five points.

THE IF-ELSE STATEMENT

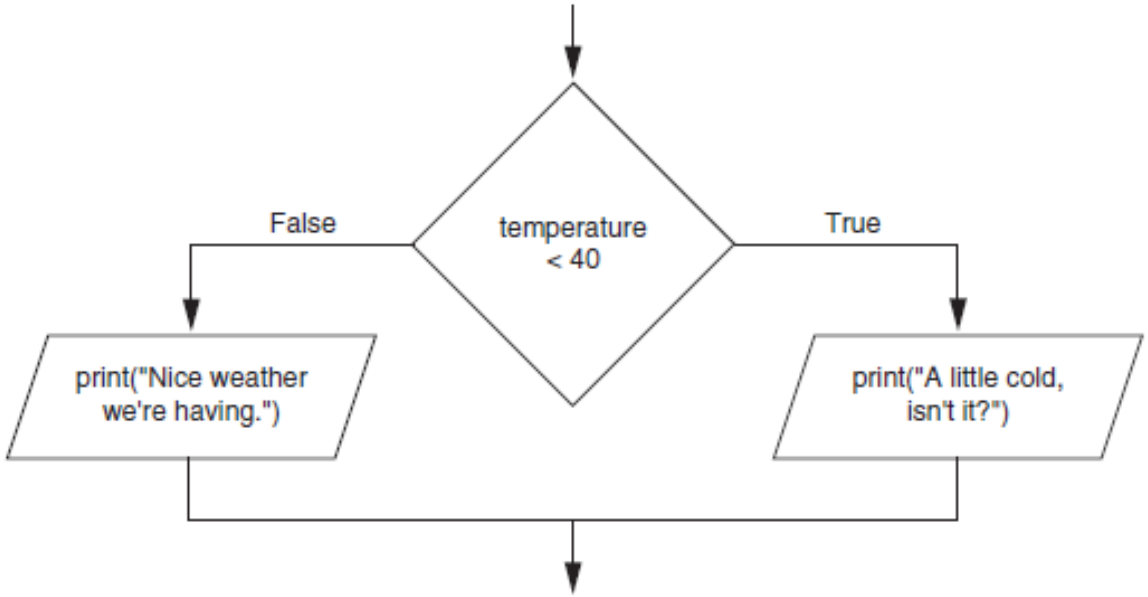
- Dual alternative decision structure: two possible paths of execution
  - One is taken if the condition is true, and the other if the condition is false
- Syntax:

```
if condition:
        statements
else:
        other statements
```

- `if` clause and `else` clause must be aligned
- Statements must be consistently indented

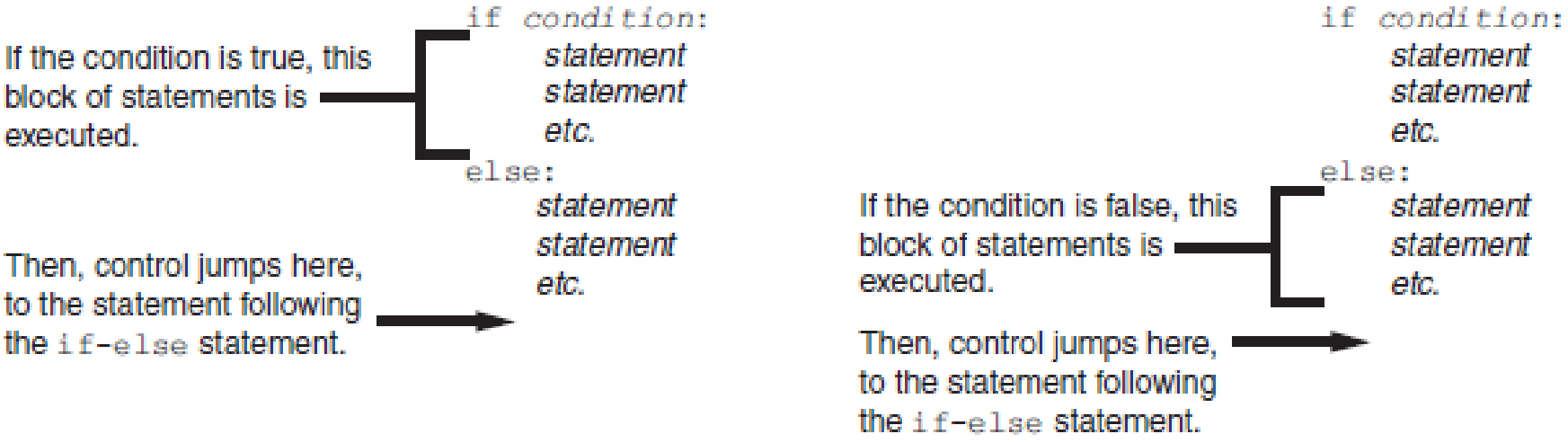
THE IF-ELSE STATEMENT (CONT'D.)

Figure 3-5 A dual alternative decision structure



THE IF-ELSE STATEMENT (CONT'D.)

Figure 3-6 Conditional execution in an `if-else` statement

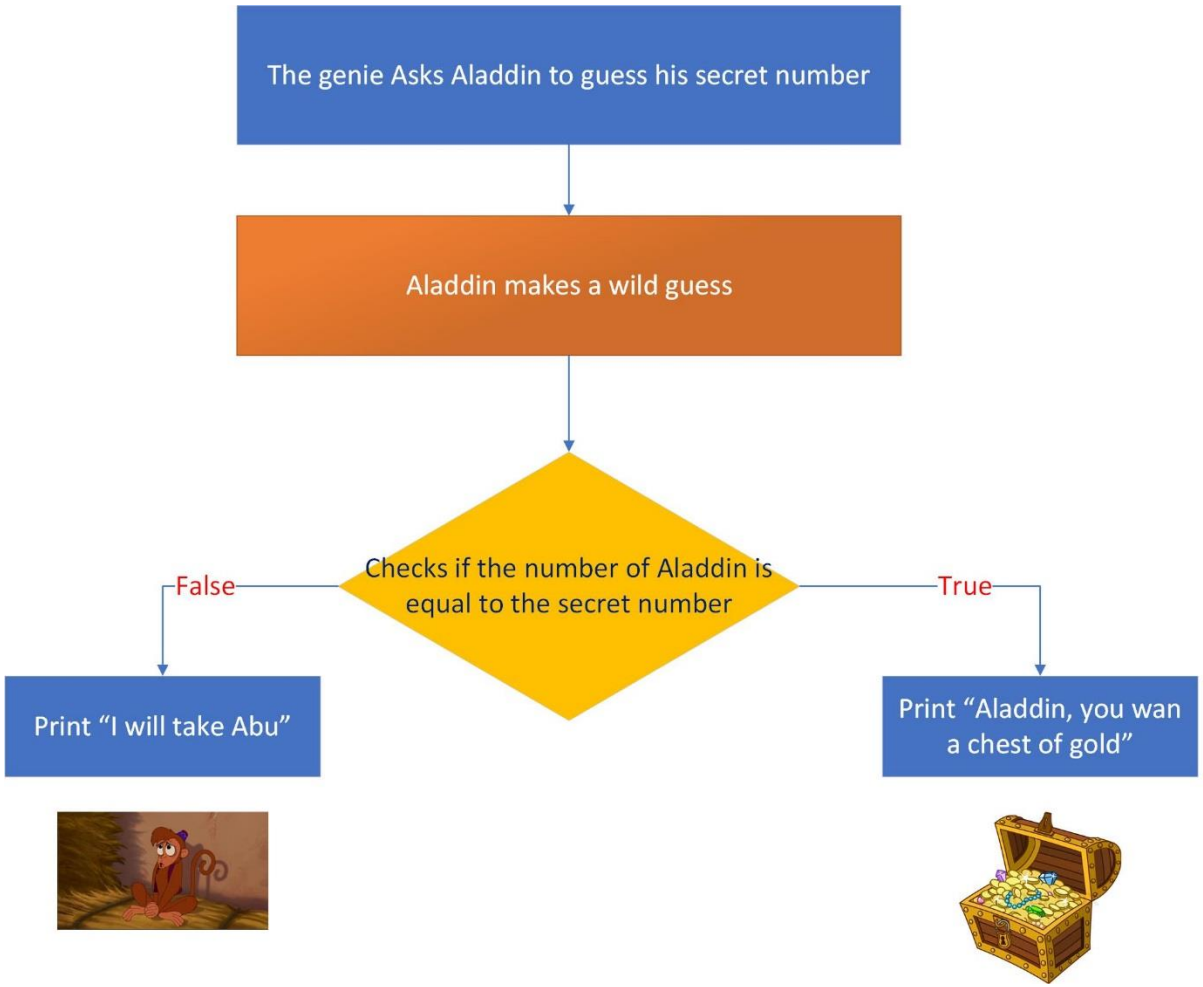


NUMBER GUESS GAME

- The genie writes a secret number on a paper.
- If Aladdin guesses the number he wins a chest of gold.
- Otherwise (else), the genie will take Abu, the poor monkey.



DFD OF THE SECRET NUMBER GUESS GAME



STEPS OF THE PROGRAM

- Store the genie's secret number in a variable
- Print a message to Aladdin (the player) Guess my secret number
- Read the user’s guess.
- Compare the user guess with the secret number.
- If the two numbers are equal then print You wan a chest of gold
- Otherwise (else) print I wan Abu

LETS WRITE THE PYTHON CODE

CARD GAME

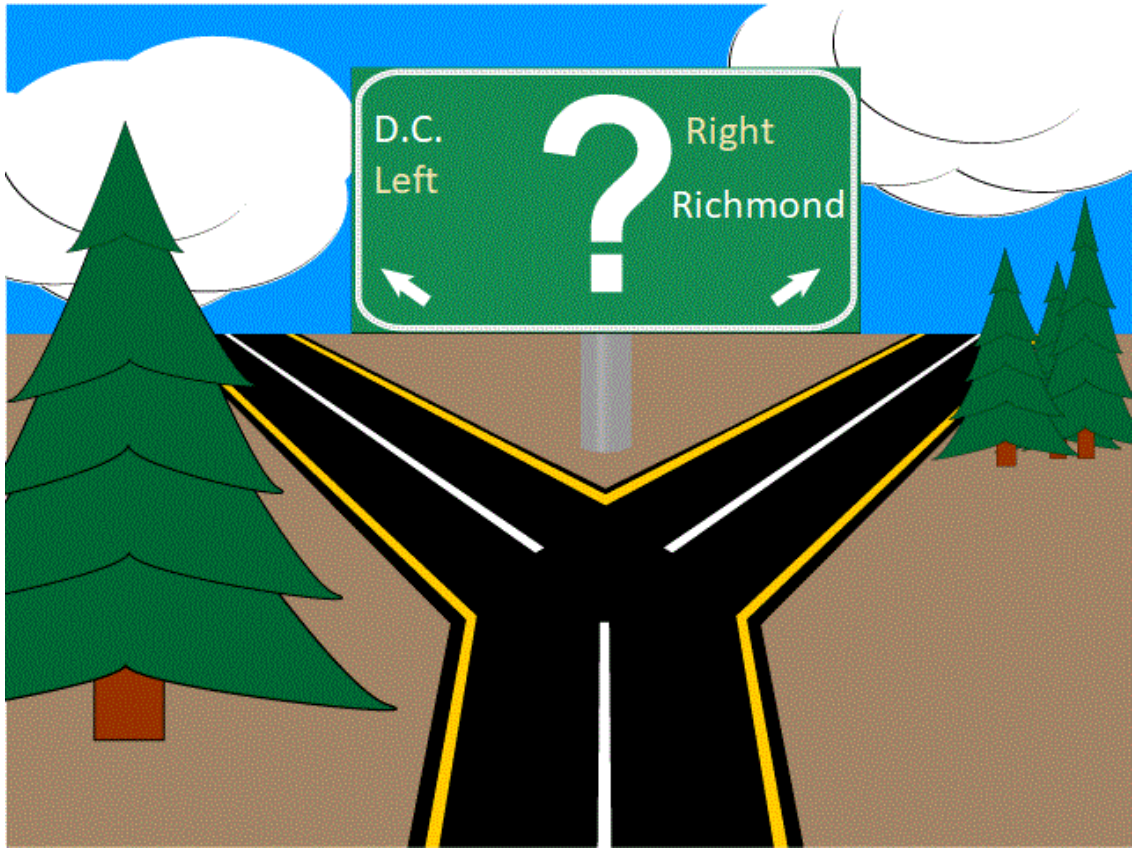
```
If (CARD is RED)
        Award YOUR team 1 point

Else
        Award OTHER team 1 point
```

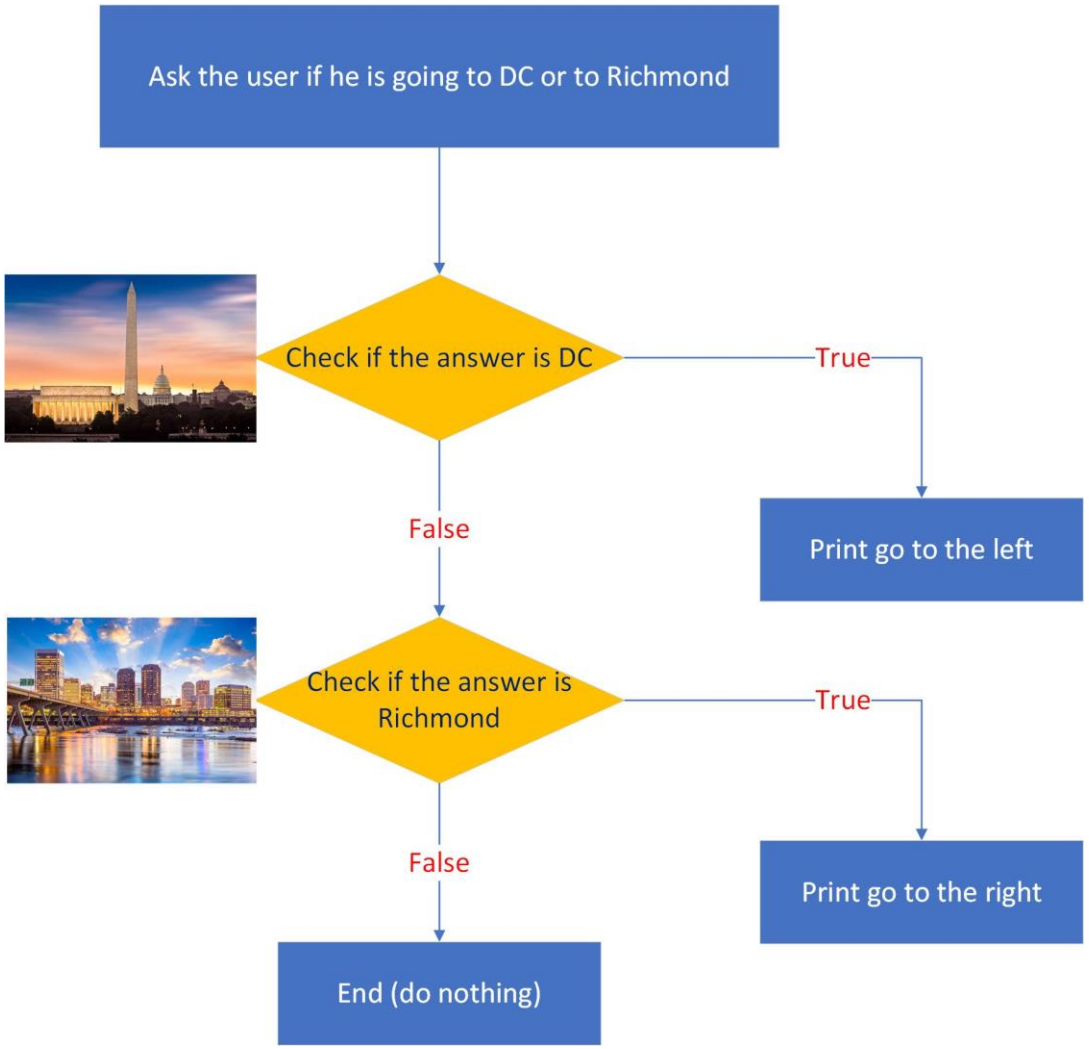
IF ELIF

- In this case we have two mutually exclusive situations.
  - Go to the left or go to the right.
  - Increase the temperature or decrease it.
  - Withdraw money or deposit money into your bank account.

GOING TO DC OR RICHMOND?



DATA FLOW DIAGRAM



DO WE REALLY NEED TO CHECK IF THE DESTINATION IS RICHMOND?

PROGRAM STEPS

- Ask the user if he want to go to DC or Richmond.
- Get the answer of the user and store it in a variable.
- If the value of the variable is DC
- Display go left
- Else if the variable is Richmond
- Then display go right

LETS WRITE THE COED ...

WHY WE NEED ELIF INSTEAD OF IF

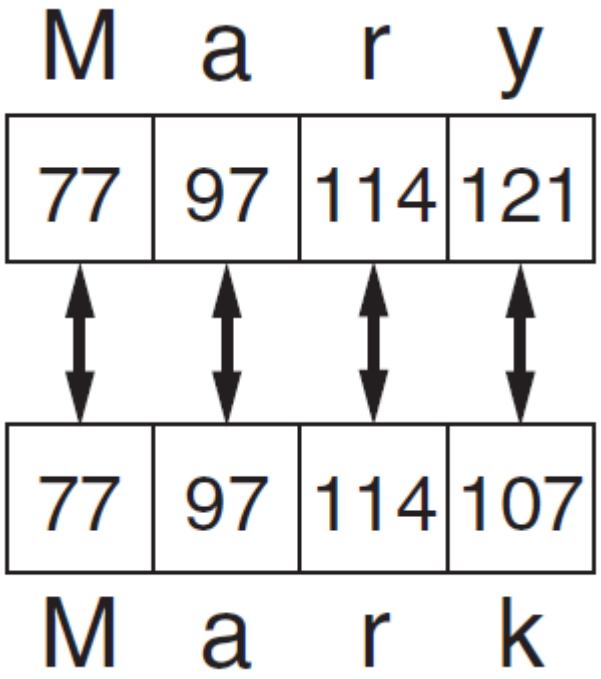
- Please note that going to the left and going to the right are mutually exclusive.
  - We cannot go in both directions
- That is why we should use elif (else if) in the second statement.

COMPARING STRINGS

- Strings can be compared using the == and != operators
- String comparisons are case sensitive
- Strings can be compared using >, <, >=, and <=
  - Compared character by character based on the ASCII values for each character
  - If shorter word is substring of longer word, longer word is greater than shorter word

COMPARING STRINGS (CONT'D.)

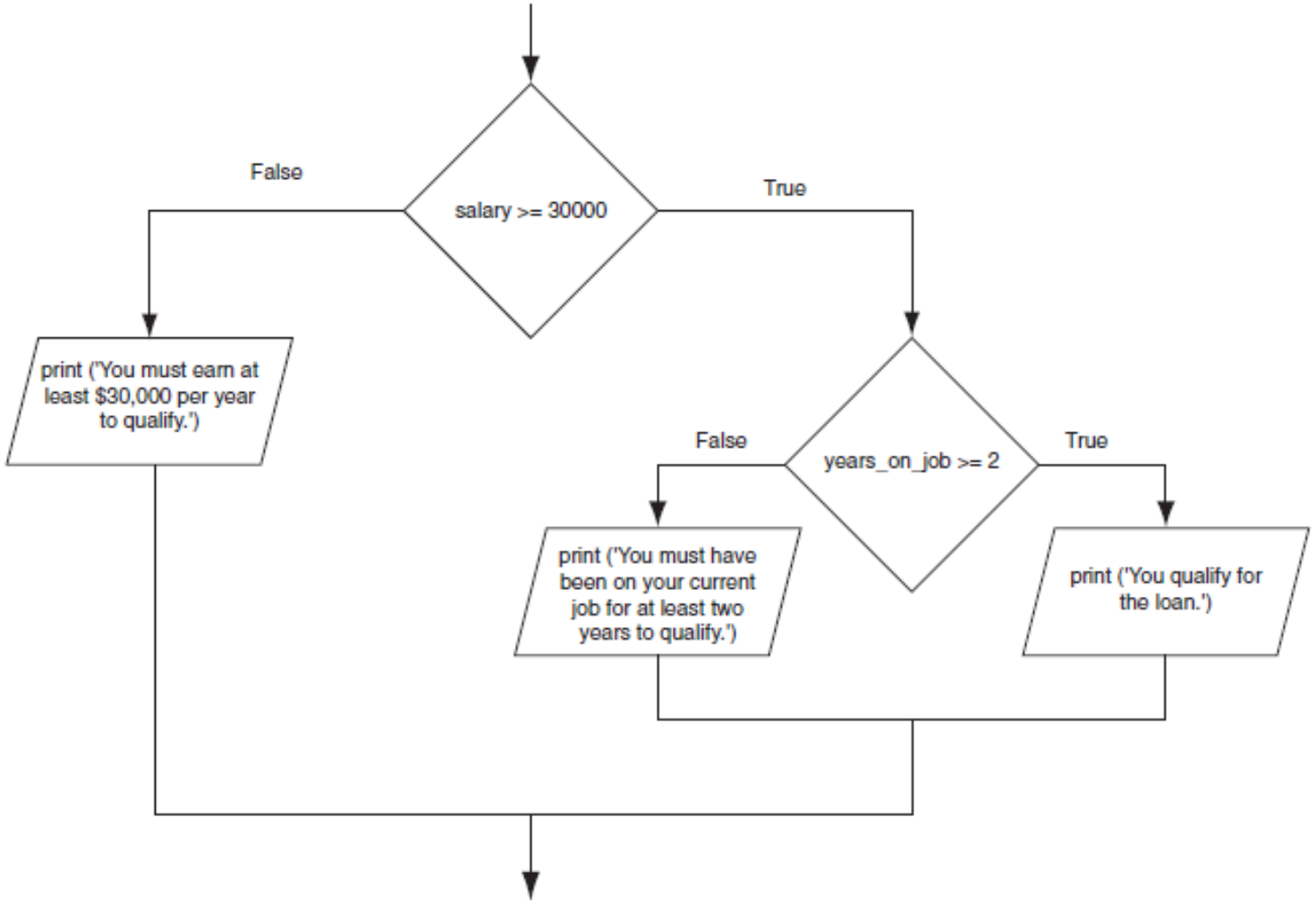
Figure 3-9 Comparing each character in a string



NESTED DECISION STRUCTURES AND THE IF-ELIF-ELSE STATEMENT

- A decision structure can be nested inside another decision structure
- Commonly needed in programs
- Example:
- Determine if someone qualifies for a loan, they must meet two conditions:
- Must earn at least $30,000/year
- Must have been employed for at least two years
- Check first condition, and if it is true, check second condition

Figure 3-12 A nested decision structure



NESTED DECISION STRUCTURES AND THE IF-ELIF-ELSE STATEMENT (CONT'D.)

- Important to use proper indentation in a nested decision structure
  - Important for Python interpreter
  - Makes code more readable for programmer
  - Rules for writing nested if statements:
    - `else` clause should align with matching `if` clause
    - Statements in each block must be consistently indented

THE IF-ELIF-ELSE STATEMENT

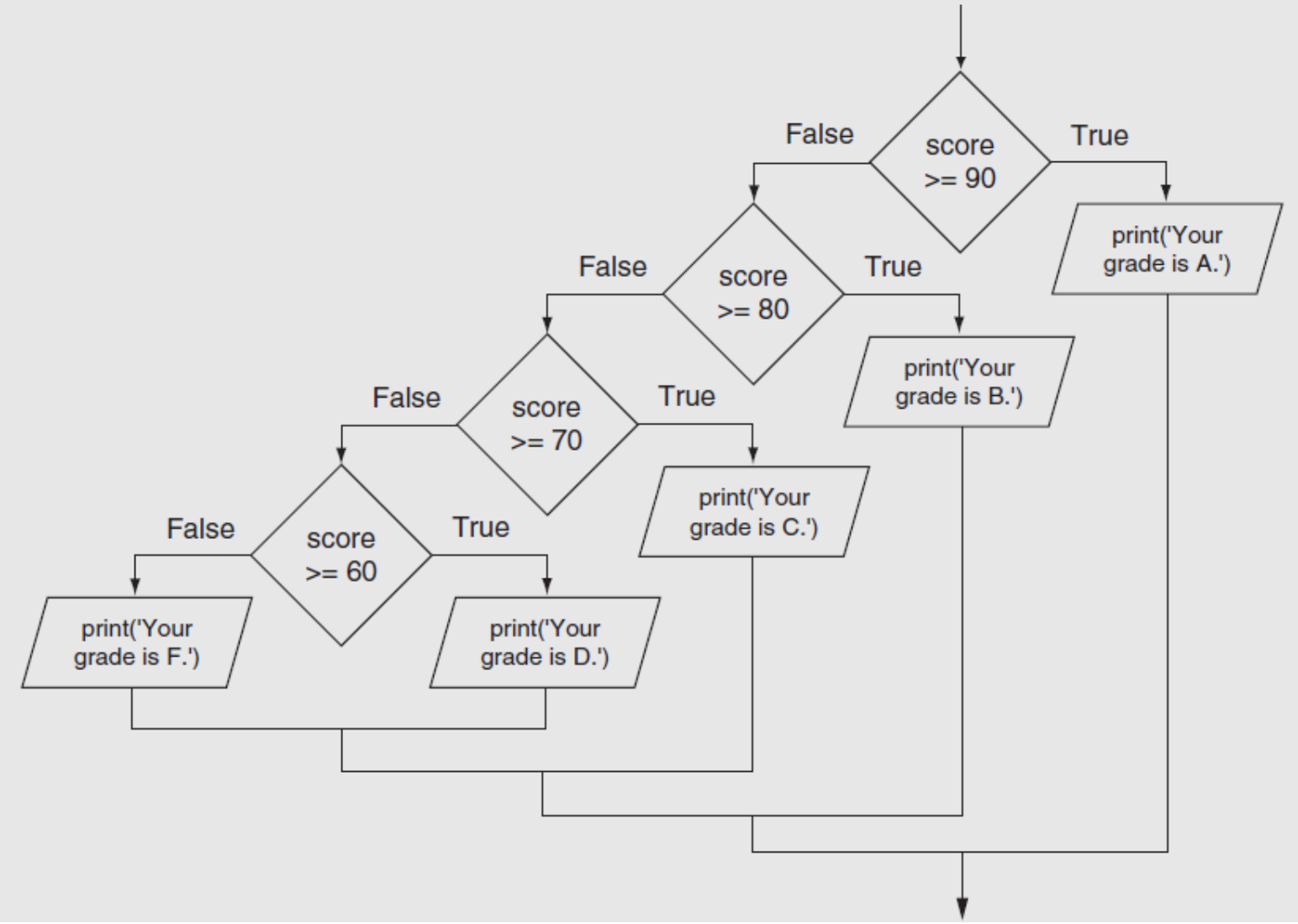
- `if-elif-else` statement: special version of a decision structure
  - Makes logic of nested decision structures simpler to write
- Can include multiple `elif` statements
- Syntax:

```
if condition1
      statements
elif condition2
      statements
else
      statements
```

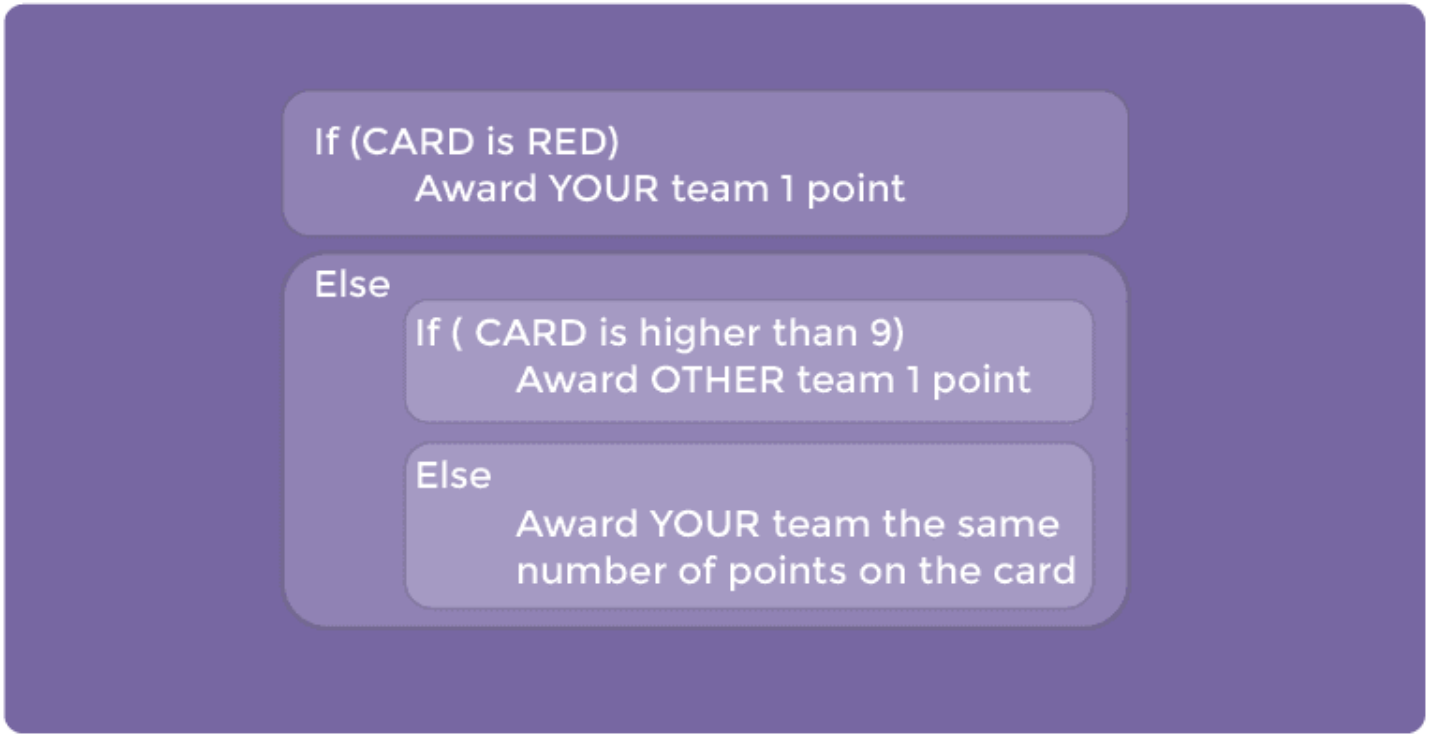
THE IF-ELIF-ELSE STATEMENT (CONT'D.)

- Alignment used with `if-elif-else` statement:
  - if, elif, and else clauses are all aligned
  - Conditionally executed blocks are consistently indented
- `if-elif-else` statement is never required, but logic easier to follow
  - Can be accomplished by nested if-else
  - Code can become complex, and indentation can cause problematic long lines

Figure 3-15 Nested decision structure to determine a grade

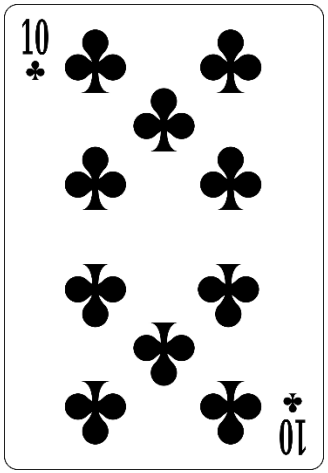


CARD EXAMPLE



EXAMPLE 1

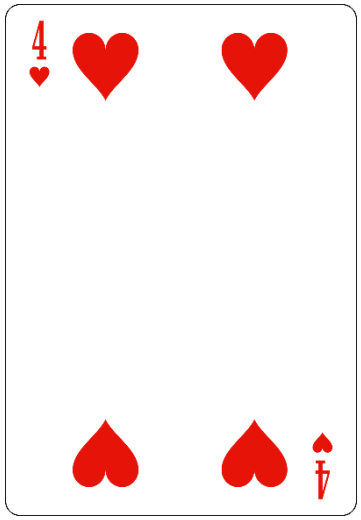
- myTeam=0



otherTeam=1

EXAMPLE 2

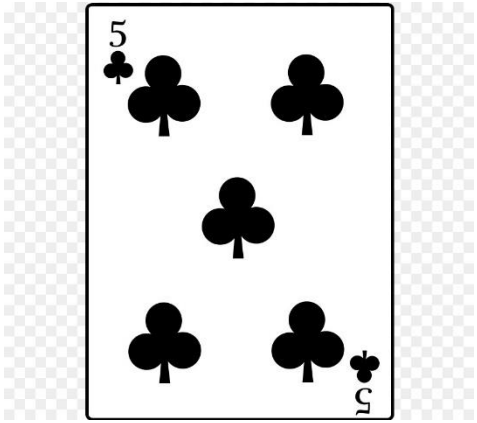
- myTeam=1



otherTeam=0

EXAMPLE 3

- myTeam=5



otherTeam=0

DFD

- Draw the DFD

LETS WRITE THE PYTHON CODE

LOGICAL OPERATORS

- Logical operators: operators that can be used to create complex Boolean expressions
- `and` operator and `or` operator: binary operators, connect two Boolean expressions into a compound Boolean expression
- `not` operator: unary operator, reverses the truth of its Boolean operand

THE AND OPERATOR

- Takes two Boolean expressions as operands
- Creates compound Boolean expression that is true only when both sub expressions are true
- Can be used to simplify nested decision structures
- Truth table for the `and` operator

Expression	Value of the Expression
false and false	false
false and true	false
true and false	false
true and true	true

THE OR OPERATOR

- Takes two Boolean expressions as operands
- Creates compound Boolean expression that is true when either of the sub expressions is true
- Can be used to simplify nested decision structures

- Truth table for the `or` operator

Expression	Value of the Expression
false or false	false
false or true	true
true or false	true
true or true	true

LETS WRITE THE CODE NOW ...

SHORT-CIRCUIT EVALUATION

- Short circuit evaluation: deciding the value of a compound Boolean expression after evaluating only one sub expression
- Performed by the `or` and `and` operators
- For `or` operator: If left operand is true, compound expression is true. Otherwise, evaluate right operand
- For `and` operator: If left operand is false, compound expression is false. Otherwise, evaluate right operand

THE NOT OPERATOR

- Takes one Boolean expressions as operand and reverses its logical value
  - Sometimes it may be necessary to place parentheses around an expression to clarify to what you are applying the not operator
- Truth table for the `not` operator

Expression	Value of the Expression
true	false
false	true

CHECKING NUMERIC RANGES WITH LOGICAL OPERATORS

- To determine whether a numeric value is within a specific range of values, use `and`
- Example: x >= 10 and x <= 20

- To determine whether a numeric value is outside of a specific range of values, use `or`
- Example: x < 10 or x > 20

BOOLEAN VARIABLES

- Boolean variable: references one of two values, `True` or `False`
- Represented by `bool` data type

- Commonly used as flags
- Flag: variable that signals when some condition exists in a program
- Flag set to `False` → condition does not exist
- Flag set to `True` → condition exists

SUMMARY

- This chapter covered:
  - Decision structures, including:
    - Single alternative decision structures
    - Dual alternative decision structures
    - Nested decision structures
  - Relational operators and logical operators as used in creating Boolean expressions
  - String comparison as used in creating Boolean expressions
  - Boolean variables